

the matrix of change of matrix basis

! Seminar 11 (10.12.2025)

$T_{BB'}$ → on columns we write the vectors in B as a lin. comb. of vectors of B' .

$$T_{BB'} = T_{B'B}^{-1}$$

$$[u]_B = T_{BB'} \cdot [u]_{B'}$$

$$f\text{-endomorphism} \Rightarrow [f]_B = T_{BB'} \cdot [f]_{B'} \cdot T_{B'B} = T_{B'B}^{-1} \cdot [f]_{B'} \cdot T_{B'B}.$$

The characteristic polynomial :

$$p(\lambda) = \det \left(\underbrace{[f]_E - \lambda I_m}_A \right)$$

Eigenvalues of f are the solutions : $p(\lambda) = 0$.

Eigenvectors of f are given by : $A \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = 0_m.$

①. $\mathbb{R}^3$

$B = (v_1, v_2, v_3) = ((1, 0, 1), (0, 1, 1), (1, 1, 1))$ - basis

$B' = (v'_1, v'_2, v'_3) = ((1, 1, 0), (-1, 0, 0), (0, 0, 1))$ - basis.

$$T_{BB'}, T_{B'B} = ?$$

Compute coord. of $u = (2, 0, -1)$ in B & B' .

$$v_1 = \alpha v'_1 + \beta v'_2 + \gamma v'_3 \Rightarrow (1, 0, 1) = \alpha(1, 1, 0) + \beta(-1, 0, 0) + \gamma(0, 0, 1).$$

$$(1, 0, 1) = (\alpha - \beta, \alpha, \gamma) \Rightarrow \begin{cases} \alpha - \beta = 1 \\ \alpha = 0 \\ \gamma = 1. \end{cases} \Rightarrow \underline{\underline{\beta = -1.}}$$

$$v_2 = (\alpha - \beta, \alpha, \gamma) \Rightarrow (0, 1, 1) = (\alpha - \beta, \alpha, \gamma) \Rightarrow \begin{cases} \alpha - \beta = 0 \\ \alpha = 1 \\ \gamma = 1 \end{cases} \Rightarrow \underline{\underline{\beta = 1}}, \underline{\underline{\gamma = 1}}$$

$$v_3 = (\alpha - \beta, \alpha, \gamma) \Rightarrow (1, 1, 1) = (\alpha - \beta, \alpha, \gamma) \Rightarrow \begin{cases} \alpha - \beta = 1 \\ \alpha = 1 \\ \gamma = 1 \end{cases} \Rightarrow \underline{\underline{\beta = 0}}, \underline{\underline{\gamma = 1}}$$

$$\Rightarrow T_{BB'} = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

①

$T_{B'B}$ like this or compute $T_{B'B}^{-1} = \frac{1}{\det T_{B'B}} \cdot T_{B'B}^*$.

$$N_1 = \alpha N_1' + \beta N_2' + \gamma N_3' = \alpha(1,1,0) + \beta(-1,0,1)$$

$$N_1' = \alpha N_1 + \beta N_2 + \gamma N_3 = \alpha(1,0,1) + \beta(0,1,1) + \gamma(1,1,1)$$

$$\Rightarrow (1,1,0) = (\alpha + \gamma, \beta + \gamma, \alpha + \beta + \gamma)$$

$$\Rightarrow \begin{cases} \alpha + \gamma = 1 \\ \beta + \gamma = 1 \\ \alpha + \beta + \gamma = 0 \end{cases}$$

$$\det T_{B'B} = \begin{vmatrix} 0 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \\ -1 & 1 & 0 \end{vmatrix} = \cancel{-1} + \cancel{1} = 1$$

$$T_{B'B}^* = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \quad T_{B'B}^* = \begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & -1 \\ -2 & 1 & 1 \end{bmatrix}$$

$$a_{11}^* = (-1)^2 \cdot \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1$$

$$a_{12}^* = (-1)^3 \cdot \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$$

$$a_{13}^* = (-1)^4 \cdot \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1$$

$$a_{21}^* = (-1)^3 \cdot \begin{vmatrix} -1 & 1 \\ 0 & 1 \end{vmatrix} = 1$$

$$a_{22}^* = (-1)^4 \cdot \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1$$

$$a_{23}^* = (-1)^5 \cdot \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} = -1$$

$$a_{31}^* = (-1)^4 \cdot \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -2$$

$$a_{32}^* = (-1)^5 \cdot \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1$$

$$a_{33}^* = (-1)^6 \cdot \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} = 1$$

$$T_{B'B}^{-1} = -1 \cdot \begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & -1 \\ -2 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & 1 \\ -1 & 1 & 1 \\ 2 & -1 & -1 \end{bmatrix}$$

$$\begin{aligned} u &= (2, 0, -1) = \alpha_1 N_1 + \alpha_2 N_2 + \alpha_3 N_3 \\ &= \alpha_1(1, 0, 1) + \alpha_2(0, 1, 1) + \alpha_3(1, 1, 1) \\ &= (\alpha_1 + \alpha_3, \alpha_2 + \alpha_3, \alpha_1 + \alpha_2 + \alpha_3) \end{aligned}$$

$$\Rightarrow \begin{cases} \alpha_1 + \alpha_3 = 2 \\ \alpha_2 + \alpha_3 = 0 \\ \alpha_1 + \alpha_2 + \alpha_3 = -1 \end{cases} \Rightarrow \alpha_1 - \alpha_2 = 2$$

$$\alpha_3 = -1 - 2 = -3$$

$$\alpha_1 = 2 + 3 = 5$$

$$\alpha_2 = -3$$

$$\Rightarrow [u]_B = \begin{bmatrix} 5 \\ -3 \\ -3 \end{bmatrix}$$

(2)

$$(2, 0, 1) = \alpha_1 (1, 1, 0) + \alpha_2 (-1, 0, 0) + \alpha_3 (0, 0, 1) \dots \Rightarrow \begin{matrix} \alpha_1 = 0 \\ \alpha_2 = -2 \\ \alpha_3 = 1 \end{matrix} \Rightarrow [u]_{B'} = \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix}$$

or using $[u]_B = T_{BB'} \cdot [u]_{B'}$

The inverse could be computed in another way!

Check: $T_{BB'} \cdot T_{B'B} = I_m$.

(2). $\mathbb{R}^2$.

$$B = (v_1, v_2) = ((1, 2), (4, 3)).$$

$$B' = (v'_1, v'_2) = ((1, 0), (2, 1))$$

$$f, g \in \text{End}_{\mathbb{R}}(\mathbb{R}^2)$$

$$[f]_B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}$$

$$[g]_{B'} = \begin{bmatrix} -7 & -13 \\ 5 & 7 \end{bmatrix}$$

$$[f+g]_B = ? = [f] + [g].$$

$$[f \circ g]_{B'} = ? = [f] \cdot [g]$$

$$v'_2 = \alpha v_1 + \beta v_2$$

$$(2, 1) = (\alpha + \beta, 2\alpha + 3\beta)$$

$$\Rightarrow \begin{cases} \alpha + \beta = 2 \\ 2\alpha + 3\beta = 1 \end{cases} \Rightarrow \alpha = 2 - \beta$$

$$2(2 - \beta) + 3\beta = 1$$

$$4 - 2\beta + 3\beta = 1$$

$$4 + \beta = 1$$

$$\beta = -3$$

$$\alpha = 2 + 3 = 5$$

$$\Rightarrow T_{B'B} = \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix}$$

(1):

$$[f]_{B'} = \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix} \cdot [f]_B \cdot \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix}$$

$$\frac{74}{148} \cdot \frac{46}{138} \cdot \frac{11}{142} \cdot \frac{46}{230} \cdot \frac{11}{213} \cdot \frac{35}{24}$$

$$[f]_B = T_{BB'} \cdot [f]_{B'} \cdot T_{B'B} \quad (1)$$

$$v_1 = \alpha v'_1 + \beta v'_2$$

$$(1, 2) = \alpha (1, 0) + \beta (2, 1) = (\alpha + 2\beta, \beta)$$

$$\begin{cases} 1 = \alpha + 2\beta \\ 2 = \beta \end{cases} \Rightarrow 1 = \alpha + 4 \Rightarrow \alpha = -3$$

$$v_2 = \alpha v'_1 + \beta v'_2 \Rightarrow \begin{cases} 1 = \alpha + 2\beta \\ 3 = \beta \end{cases} \Rightarrow \alpha = 1 - 6 = -5$$

$$\Rightarrow T_{BB'} = \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix}$$

$$v'_1 = \alpha v_1 + \beta v_2 \Rightarrow (1, 0) = \alpha (1, 2) + \beta (4, 3)$$

$$\Rightarrow (1, 0) = (\alpha + 4\beta, 2\alpha + 3\beta)$$

$$\Rightarrow \begin{cases} \alpha + 4\beta = 1 \\ 2\alpha + 3\beta = 0 \end{cases} \Rightarrow \alpha = 1 - 4\beta$$

$$2(1 - 4\beta) + 3\beta = 0 \Rightarrow 2 - 8\beta + 3\beta = 0$$

$$2 - 5\beta = 0 \Rightarrow 5\beta = 2 \Rightarrow \beta = \frac{2}{5}$$

$$\alpha = 1 - 4 \cdot \frac{2}{5} = 1 - \frac{8}{5} = -\frac{3}{5}$$

$$[f+g]_B = [f]_B + [g]_B = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} + \begin{bmatrix} -20 & -32 \\ 13 & 20 \end{bmatrix} = \begin{bmatrix} -19 & -30 \\ 12 & 19 \end{bmatrix}$$

$$I \text{ need } [g]_B = T_{BB'} \cdot [g]_{B'} \cdot T_{B'B}$$

$$= \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} -7 & -13 \\ 5 & 7 \end{bmatrix} \cdot \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -21 - 25 & -39 - 35 \\ -14 + 15 & -26 + 21 \end{bmatrix} \cdot \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -46 & -74 \\ +1 & -5 \end{bmatrix} \cdot \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -138 + 148 & -230 + 213 \\ 3 + 10 & 5 + 15 \end{bmatrix} = \begin{bmatrix} -20 & -32 \\ 13 & 20 \end{bmatrix} \quad (3)$$

$$[f]_{B'} = \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} 8 & 11 \\ -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 5 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} 24-22 & 40-3 \\ -5 & -8 \end{bmatrix} \\ = \begin{bmatrix} 2 & 7 \\ -5 & -8 \end{bmatrix} \cdot \begin{bmatrix} 8 & 13 \\ -5 & -8 \end{bmatrix}$$

$$[f \circ g]_{B'} = [f]_{B'} \cdot [g]_{B'} = \begin{bmatrix} 2 & 7 \\ -5 & -8 \end{bmatrix} \cdot \begin{bmatrix} -7 & -13 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} -14+35 & -26+49 \\ 35-40 & 65-56 \end{bmatrix} =$$

$$= \begin{bmatrix} 21 & 23 \\ -5 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 13 \\ -5 & -8 \end{bmatrix} \cdot \begin{bmatrix} -7 & -13 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} -56+65 & -104+91 \\ 35-40 & 65-56 \end{bmatrix} = \begin{bmatrix} 9 & -13 \\ -5 & 9 \end{bmatrix} \xrightarrow{\frac{13}{9}}$$

④ $f \in \text{End}_{\mathbb{R}}(\mathbb{R}^2)$

$$f(x, y) = (3x+3y, 2x+4y)$$

$$p(\lambda) = \det([f]_E - \lambda I_m) = 0.$$

a) Det. eigenvalues, eigenvectors of f

b) Write a basis B of eigenvectors of f .

$$A \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = 0_m.$$

$$[f]_E = \begin{bmatrix} 3 & 3 \\ 2 & 4 \end{bmatrix}$$

$$p(\lambda) = \begin{vmatrix} 3-\lambda & 3 \\ 2 & 4-\lambda \end{vmatrix} = (3-\lambda)(4-\lambda) - 6 = 12 - 7\lambda + \lambda^2 - 6 \\ = \lambda^2 - 7\lambda + 6 = 0$$

$$\cancel{\lambda^2 - 6\lambda - \lambda + 6 = 0}$$

$$\lambda^2 - 6\lambda - \lambda + 6 = 0$$

$$\lambda(\lambda-6) - (\lambda-6) = 0$$

$$(\lambda-6)(\lambda-1) = 0 \quad \begin{cases} \lambda_2 = 6 \\ \lambda_1 = 1 \end{cases} \text{ eigenvalues.}$$

$$\begin{bmatrix} 3-\lambda & 3 \\ 2 & 4-\lambda \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} (3-\lambda)x_1 + 3x_2 = 0 \\ 2x_1 + (4-\lambda)x_2 = 0 \end{cases}$$

Case I: $\lambda_1 = 1$

$$\Rightarrow \begin{cases} 2x_1 + 3x_2 = 0 \\ 2x_1 + 3x_2 = 0. \end{cases} \Rightarrow N_1 = \begin{bmatrix} -\frac{3}{2} \\ 1 \end{bmatrix} \Rightarrow N_1(\lambda_1) = \left\langle \left(-\frac{3}{2}, 1\right) \right\rangle$$

Case II: $\lambda_2 = 6$

$$\Rightarrow \begin{cases} -3x_1 + 3x_2 = 0 \\ 2x_1 - 2x_2 = 0 \end{cases} \Rightarrow \begin{cases} -x_1 + x_2 = 0 \\ -x_1 + x_2 = 0. \end{cases} \Rightarrow N_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow N_2(\lambda_2) = \langle (1, 1) \rangle$$

$$\Rightarrow \nexists \alpha \text{ s.t. } v(\lambda_2) = \alpha v(\lambda_1)$$

Lim. indep: $\begin{vmatrix} 1 & 1 \\ -\frac{3}{2} & 1 \end{vmatrix} = 1 + \frac{3}{2} = \frac{5}{2} \neq 0$

?

$$\Rightarrow B = \left\langle \left(1, 1\right), \left(-\frac{3}{2}, 1\right) \right\rangle.$$

9) $A \in M_2(\mathbb{R})$.

λ_1, λ_2 eigenvalues of A in $\mathbb{C}$.

a) $\lambda_1 + \lambda_2 = \text{Tr}(A)$.

b) $\lambda_1 \cdot \lambda_2 = \det(A)$

c) A has all eigenvalues in $\mathbb{R}$ $\Leftrightarrow (\text{Tr}(A))^2 - 4 \det(A) \geq 0$.

a) $p(\lambda) = \lambda^2 - \lambda(a+d) + ad - bc$ (1)

$$p(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - \lambda\lambda_2 - \lambda_1\lambda + \lambda_1\lambda_2 = \lambda^2 - \lambda(\lambda_2 + \lambda_1) + \lambda_1\lambda_2$$

b) $\Rightarrow \begin{cases} a+d = \lambda_1 + \lambda_2 = \text{Tr}(A) \\ ad - bc = \lambda_1 \cdot \lambda_2 = \det(A) \end{cases}$

"Prove... using that the $\text{Tr}(A) = \lambda_1 + \lambda_2$ and $\det \dots$ ".

c) $p(\lambda) \stackrel{(1)}{=} \lambda^2 - \lambda \cdot \text{Tr}(A) + \det(A)$

$$p(\lambda) = 0 \Rightarrow \Delta = (-\text{Tr}(A))^2 - 4 \cdot 1 \cdot \det(A)$$

If $\Delta \geq 0 \Rightarrow \exists \lambda_1, \lambda_2 \in \mathbb{R}$. not necessarily equal
 $\downarrow$
eigenvalues